

## Assignment 6: Black Holes

*to be handed in online by Thursday, July 16th*

Students taking the “[...] Black Holes” physics course are only expected to submit the exercises marked with an asterisk\*, but are obviously invited to try all of them. Students taking the “[...] Classical Systems” maths course are expected to submit (almost) all exercises.

If you skip an exercise, be sure to read it at least, since you might need its result later.

We have hidden most hints as white text, to encourage you to try to solve the problems without looking at them. If you are stuck, simply copy the white text somewhere else to reveal it.

Feel free to use **Mathematica**, but submit the notebook or screenshots of it.

### Exercise 0: Evaluation

Use the QR codes below to participate in the evaluation. The links are also on WueCampus.



(a) Geo&Ana QR Code



(b) BHs QR Code

Figure 1: Evaluation QR Codes. Fill yours! :)

### Exercise 1: The Reissner-Nordström Black Hole

The Reissner-Nordström solution to the Einstein-Maxwell equations reads

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (1)$$

$$A = -\frac{Q}{4\pi r}dt - \frac{P}{4\pi} \cos\theta d\phi, \quad (2)$$

where

$$f(r) = 1 - \frac{2G_N M}{r} + \frac{Q^2 + P^2}{4\pi r^2}. \quad (3)$$

**a)\*** Show that

$$Q = \int_{S_\infty^2} \star F \quad \text{and} \quad P = \int_{S_\infty^2} F, \quad (4)$$

where  $S_\infty^2$  is a 2-sphere at  $r = \infty$  on a surface of constant  $t$ , and  $F = dA$ .

What is the physical interpretation of  $Q$  and  $P$ ?

## Exercise 2: Penrose Process

**a)** Prove that, if a vector field  $\xi$  preserves the Maxwell field  $F = dA$  (i.e.  $\mathcal{L}_\xi F = 0$ ), then locally there exists a scalar potential  $\Phi$  such that  $\iota_\xi F = d\Phi$ .

*Hint:*

**b)** The equation of motion of a particle of charge  $q$  and 4-velocity  $u^\mu$  is  $qF^\mu{}_\nu u^\nu = mu^\nu \nabla_\nu u^\mu$ . Now, let  $\xi$  be a Killing vector field that preserves the Maxwell field. Show that

$$K = m\xi_\mu u^\mu - q\Phi \quad (5)$$

is conserved along the particle's worldline,  $\dot{K} = 0$ .

**c)** Deduce that, for a particle of mass  $m$  moving in the equatorial plane of a Reissner-Nordstrom black hole with  $Q > 0$  and  $P = 0$ , the quantities

$$\mathcal{E} = f(r)\dot{t} + \frac{qQ}{4\pi mr}, \quad \text{and} \quad h = r^2\dot{\phi} \quad (6)$$

are constant, where the dot denotes derivative with respect to the proper time  $\tau$  of the particle. Hence show that the motion in the radial direction is determined by the equation

$$\dot{r}^2 + f(r) \left(1 + \frac{h^2}{r^2}\right) = \left(\mathcal{E} - \frac{qQ}{4\pi mr}\right)^2. \quad (7)$$

*Hint:*

**d)\*** *The Penrose Process.* A particle  $P_1$ , with  $q_1 > 0$ , falls from  $r = \infty$  towards a Reissner-Nordstrom black hole with  $Q > 0$  and  $P = 0$ . Just before it crosses the event horizon, it decays into two other particles  $P_2$  and  $P_3$ , where  $P_2$  has charge  $q_2 < 0$ . The decay happens such that  $P_2$  initially has  $\dot{r} \simeq 0$ , and then this particle falls into the black hole, while  $P_3$  escapes to  $r = \infty$ .

Let  $E_i = m_i \mathcal{E}_i$  denote the energy of  $P_i$ . Show that  $E_1 > 0$  and  $E_2 < 0$ .

Hence, by charge conservation  $E_3 > E_1$ , that is, the particle returning to infinity has more energy than the initial particle! This is consistent because  $P_2$  has carried negative energy into the black hole. Hence energy (and charge) are extracted from the black hole in this process.

**e)\*** Read Reall's Section 7.5 to learn about the Penrose process in Kerr black holes.

### Exercise 3: Penrose Diagram of de Sitter Space

In so-called *global coordinates* the metric of  $(3 + 1)$ -dimensional de Sitter space reads

$$ds^2 = -d\tau^2 + R^2 \cosh^2\left(\frac{\tau}{R}\right) d\Omega_3^2, \quad \tau \in (-\infty, +\infty). \quad (8)$$

As is evident from this expression, in the far future and far past dS spacetime can be viewed as a three-dimensional sphere,  $S^3$  that asymptotically expands exponentially fast.

**a)\*** Find a *conformal time coordinate*  $\eta$  such that the metric takes the form

$$ds^2 = \frac{R^2}{\cos^2(\eta)} (-d\eta^2 + d\Omega_3^2). \quad (9)$$

**b)** Argue that the line element of a  $d + 1$ -dimensional sphere can always be written as

$$d\Omega_{d+1}^2 = d\chi^2 + \sin^2 \chi d\Omega_d^2. \quad (10)$$

**c)\*** Use eq. (10) in eq. (9) and draw the Penrose diagram of de Sitter where, as usual, a two-sphere is suppressed. Make sure to label your axes and give the coordinate ranges. Use the Penrose diagram to argue that observers sitting on opposite points of the  $S^3$  cannot communicate.

**d)\*** We define the event horizon as the boundary of the set of points with which an observer at  $\chi = 0$  can communicate (both receive and send signals). Draw the event horizon into the Penrose diagram and show that it is a Killing horizon with KVF

$$\xi = \frac{1}{R}(\cos \chi \cos \eta \partial_\eta - \sin \chi \sin \eta \partial_\chi). \quad (11)$$

**e)\*** The region with which this observer at rest can communicate can be expressed in *static patch coordinates*

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega_2^2 \quad (12)$$

with

$$f(r) = 1 - \frac{r^2}{R^2} \quad (13)$$

using the coordinate change (you do not need to show this)

$$\frac{r}{R} = \frac{\sin \chi}{\cos \eta}, \quad (14)$$

$$\tanh\left(\frac{t}{R}\right) = \frac{\cos \chi}{\sin \eta}. \quad (15)$$

Rewrite eq. (11) in static patch coordinates and compute the associated surface gravity.

*Hint:*

**(Optional) Exercise 4: Proof of the 0th Law of Black Hole Mechanics**

*This is a long calculation, so we leave it as an optional exercise for all students.*

**a)** Let  $\xi$  be a Killing vector field. Show that

$$\nabla_\mu \nabla_\nu \xi_\rho = R^\sigma{}_{\mu\nu\rho} \xi_\sigma. \quad (16)$$

**b)** Let  $\mathcal{N}$  be a Killing horizon of a Killing vector field  $\xi$  with surface gravity  $\kappa$ .

If we know that  $A = 0$  on  $\mathcal{N}$  for some tensor  $A_{\mu_1 \dots \mu_i}$ , then  $A \cdot B = A_{\mu_1 \dots \mu_i} B^{\mu_1 \dots \mu_i} = 0$  on  $\mathcal{N}$  for any tensor  $B^{\mu_1 \dots \mu_i}$ . Hence  $\mathcal{N}$  is a surface of constant  $A \cdot B$  so  $d(A \cdot B)$  is normal to  $\mathcal{N}$ .

i. Show that  $\xi \wedge d(A \cdot B) = 0$  on  $\mathcal{N}$ .

ii. Show that this implies  $\xi_{[\mu} \nabla_{\nu]} A_{\sigma_1 \dots \sigma_i} = 0$  on  $\mathcal{N}$ .

iii. Taking  $A_\mu = \xi^\nu \nabla_\nu \xi_\mu - \kappa \xi_\mu$ , show that, on  $\mathcal{N}$ ,

$$\xi_\mu \xi_{[\sigma} \nabla_{\rho]} \kappa + \kappa \xi_{[\sigma} \nabla_{\rho]} \xi_\mu = (\xi_{[\sigma} \nabla_{\rho]} \xi^\nu) \nabla_\nu \xi_\mu + \xi^\nu \xi_{[\sigma} R^\lambda{}_{\rho]\nu\mu} \xi_\lambda. \quad (17)$$

**c)** Recall that Frobenius' theorem states that

$$\xi \text{ is hypersurface orthogonal} \iff \xi \wedge d\xi = 0 \iff \xi_{[\mu} \nabla_{\nu]} \xi_{\rho]} = 0. \quad (18)$$

i. Use Frobenius theorem to show that, on  $\mathcal{N}$ ,

$$\xi_\rho \nabla_\mu \xi_\nu = -2\xi_{[\mu} \nabla_{\nu]} \xi_\rho. \quad (19)$$

ii. Hence show that, on  $\mathcal{N}$ ,

$$\kappa \xi_{[\sigma} \nabla_{\rho]} \xi_\mu = (\xi_{[\sigma} \nabla_{\rho]} \xi^\nu) \nabla_\nu \xi_\mu, \quad (20)$$

so equation (17) reduces to

$$\xi_\mu \xi_{[\sigma} \nabla_{\rho]} \kappa = \xi^\nu \xi_{[\sigma} R^\lambda{}_{\rho]\nu\mu} \xi_\lambda. \quad (21)$$

**d)** Now, set  $A_{\mu\nu\rho} = \xi_\rho \nabla_\mu \xi_\nu + 2\xi_{[\mu} \nabla_{\nu]} \xi_\mu$  and use the previous results to show that, on  $\mathcal{N}$ ,

$$\xi_\rho \xi_{[\sigma} \nabla_{\lambda]} \nabla_\mu \xi_\nu = -2\xi_{[\sigma} \nabla_{\lambda]} \nabla_{[\nu} \xi_{\rho]} \xi_\mu, \quad (22)$$

and hence

$$\xi_\rho \xi_{[\sigma} R^\kappa{}_{\lambda]\mu\nu} \xi_\kappa = 2\xi_{[\sigma} R^\kappa{}_{\lambda]\rho[\nu} \xi_{\mu]} \xi_\kappa. \quad (23)$$

**e)** Contract indices  $\rho$  and  $\lambda$  to show that, on  $\mathcal{N}$ ,

$$0 = \xi_{[\mu} R_{\nu]}{}^\kappa \xi_\kappa \xi_\sigma + \xi_{[\mu} R^\kappa{}_{\nu]\rho\sigma} \xi^\rho \xi_\kappa. \quad (24)$$

so equation (21) reduces to

$$\xi_{[\sigma} \nabla_{\rho]} \kappa = -\xi_{[\sigma} R_{\rho]}{}^\kappa \xi_\kappa. \quad (25)$$

**f)** Read Reall's Section 9.3 to convince yourselves that the DEC, Raychaudhuri's equation, and the Einstein Equations imply that the RHS of equation (25) above vanishes for the (Killing) event horizons of stationary black holes. Hence  $\kappa$  is constant on the horizon.